

Answering Multi-Agent Machine Learning Assignment 2

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Question 1

Part A

This is trivial from the fact that

$$\begin{aligned}\mathbb{E}[G(x; \xi)] &= \frac{1}{d} \sum_{\xi=1}^d G(x; \xi) \\ &= \frac{1}{d} \sum_{\xi=1}^d d \left[\frac{\partial f(x)}{\partial x} \right] e_{\xi} \\ &= \sum_{\xi=1}^d \left[\frac{\partial f(x)}{\partial x} \right] e_{\xi} \\ &= \nabla f(x)\end{aligned}$$

Q.E.D.

Part B

Again, we have

$$\begin{aligned}\mathbb{E}[\|G(x; \xi)\|^2] &= \frac{1}{d} \sum_{\xi=1}^d \|G(x; \xi)\|^2 \\ &= \frac{1}{d} \sum_{\xi=1}^d d^2 \left[\frac{\partial f(x)}{\partial x} \right]^2 \\ &= d \sum_{\xi=1}^d \left[\frac{\partial f(x)}{\partial x} \right]^2 \\ &\leq dL^2\end{aligned}$$

The final step follows from the definition of L-Lipschitz smoothness.

Question 2

Part 1

For proving the above statement, let's start by writing it out.

$$\text{dist}(x, \Omega) = \inf_{w \in \Omega} \|x - w\|$$

let us consider another point y . For 1-Lipschitzness, we have

$$\|\text{dist}(x, \Omega) - \text{dist}(y, \Omega)\| = \left| \inf_{w \in \Omega} \|x - w\| - \inf_{w \in \Omega} \|y - w\| \right|$$

thus, we have to prove that,

$$\implies \left| \inf_{w \in \Omega} \|x - w\| - \inf_{w \in \Omega} \|y - w\| \right| \leq \|x - y\|$$

By triangle inequality, we have

$$\begin{aligned} \|x - w\| &\leq \|x - y\| + \|y - w\| \\ \implies \|x - w\| - \|y - w\| &\leq \|x - y\| \end{aligned}$$

and taking infimum over $w \in \Omega$, we have

$$\inf_{w \in \Omega} \|x - w\| - \inf_{w \in \Omega} \|y - w\| \leq \|x - y\|$$

Q.E.D.

Part 2

We have to prove that $\text{dist}(x, \Omega)$ is convex $\iff \Omega$ is convex.

For $(\implies)$ we know that if $\text{dist}(x, \Omega)$ is convex, then for any $x, y \in \Omega$ and $t \in [0, 1]$, we have

$$\text{dist}(tx + (1-t)y, \Omega) \leq t \cdot \text{dist}(x, \Omega) + (1-t) \cdot \text{dist}(y, \Omega) = 0$$

but $\text{dist}(tx + (1-t)y, \Omega) \geq 0$ meaning, $\text{dist}(tx + (1-t)y, \Omega) = 0$ which means $tx + (1-t)y \in \Omega$. Thus, Ω is convex.

For $(\impliedby)$ we have to prove that if Ω is convex, then $\text{dist}(x, \Omega)$ is convex. For that, we choose a contrapositive approach, where we show that if $\text{dist}(x, \Omega)$ is not convex, then Ω is not convex. We have to prove that for any x, y and $t \in [0, 1]$, if

$$\text{dist}(tx + (1-t)y, \Omega) > t \cdot \text{dist}(x, \Omega) + (1-t) \cdot \text{dist}(y, \Omega)$$

then Ω is not convex. Let us assume that $x, y \in \Omega$ now because Ω is not convex there exists $tx - (1-t)y \notin \Omega$, and for that,

$$\text{dist}(tx - (1-t)y, \Omega) > 0$$

Thus, function dist will not be convex. Q.E.D.

Question 3

Part a

Using the same inequality given in the question for convex functions, if we put

$$g = \partial f(x) = 0$$

then

$$\forall y, f(y) \geq f(x)$$

Q.E.D.

Part b

Fixing x and $g \in \partial f(x)$, and $v \in \mathbb{S}^{d-1}$ and take $t > 0$. From the inequality given,

$$\begin{aligned} f(x + tv) &\geq f(x) + \langle g, tv \rangle \\ \implies \langle g, v \rangle &\leq \frac{f(x + tv) - f(x)}{t}. \end{aligned}$$

Because f is L-Lipschitz smooth, we have

$$\langle g, v \rangle \leq \frac{L\|tv\| - 0}{t} = L$$

We get after taking the supremum over unit vectors. Q.E.D.

Question 4

Given the operator $S(x) = x - \alpha \nabla f(x)$, where f is m -strongly convex, we want to show:

$$\|S(x) - S(y)\| \leq (1 - \alpha m)\|x - y\|$$

Write the difference

$$S(x) - S(y) = (x - y) - \alpha(\nabla f(x) - \nabla f(y))$$

Using the parallelogram identity for the norm squared,

$$\|S(x) - S(y)\|^2 = \|x - y\|^2 - 2\alpha \langle \nabla f(x) - \nabla f(y), x - y \rangle + \alpha^2 \|\nabla f(x) - \nabla f(y)\|^2$$

Since f is m -strongly convex, its gradient satisfies the strong monotonicity property,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq m\|x - y\|^2$$

Since f is M -smooth, the gradient is Lipschitz continuous,

$$\|\nabla f(x) - \nabla f(y)\| \leq M\|x - y\|$$

Substitute these into the norm squared,

$$\|S(x) - S(y)\|^2 \leq \|x - y\|^2 - 2\alpha m\|x - y\|^2 + \alpha^2 M^2\|x - y\|^2$$

$$\|S(x) - S(y)\|^2 \leq (1 - 2\alpha m + \alpha^2 M^2) \|x - y\|^2$$

Taking square roots,

$$\|S(x) - S(y)\| \leq \sqrt{1 - 2\alpha m + \alpha^2 M^2} \|x - y\|$$

For small enough α ,

$$\sqrt{1 - 2\alpha m + \alpha^2 M^2} \leq 1 - \alpha m$$

so that

$$\|S(x) - S(y)\| \leq (1 - \alpha m)\|x - y\|$$